



NORTHERN UNIVERSITY

OF BUSINESS & TECHNOLOGY KHULNA

Department of Computer Science and Engineering

Project Title: AI-Based Smart Quiz System.

Course Code: CSE 4204

Course Title: Mobile Computing Lab

Submitted By:

Team Name : CSE4204-8A-T03

Section : 8A

GitHub Repository: <https://github.com/sudipto224/AI-based-Smart-Quiz-System>

Submitted to:

Md. Riaz Mahmud

Designation: Assistant Professor

Department of CSE

Date of Submission: 29/06/2026

TEAM INFORMATION

Official Team Name	CSE4204-8A-T03
Section	8A
Project Title	AI-Based Smart Quiz System
GitHub Repository	https://github.com/sudipto224/AI-based-Smart-Quiz-System

Team Information:

Role	Full Name	Student ID
Team Leader + Backend	Sudipto Das	11220320830
Frontend	Chaon Das	11210320629
Artificial Intelligence	Rakibur Rahman Mishan	11220320829
Database	Aditi Roy	11220320846

Backend Technology Stack

The AI-Based Smart Quiz System is developed using a modern and robust backend technology stack. The following technologies have been utilized to ensure scalability, security, and efficient performance throughout the development process.

- Backend Framework: Laravel 12.62.0
- Programming Language: PHP 8.2.12
- Database: MySQL (XAMPP) 10.4.32-MariaDB
- Authentication: Laravel Breeze (session-based)
- Server: Apache 2.4.58
- API Testing: Postman
- Version Control: Git & GitHub

Laravel 12 was selected as the primary framework due to its elegant syntax, built-in authentication scaffolding, and support for rapid application development. PHP 8.2.12 provides modern language features including improved type safety and performance enhancements. MySQL, running via XAMPP with MariaDB 10.4.32, serves as the relational database management system. Laravel Breeze handles authentication with a lightweight, session-based approach. The development server runs on Apache 2.4.58, and API endpoints are tested using Postman. All source code is version-controlled through Git and hosted on GitHub.

Database Design Summary

The database for the AI-Based Smart Quiz System is named `ai_smart_quiz_db`. It has been designed to support multi-role user management, course and question storage, quiz attempt tracking, and cheating detection. The database consists of four primary tables as described below.

Tables :

1. users

Stores user registration and profile data. Fields: id (PK), name, email, password, is_teacher (1 = Teacher, 0 = Student), created_at, updated_at, email_verified_at, remember_token.

2. courses

Stores course information created by teachers. Fields: id (PK), teacher_id (FK referencing users.id), name, description, time_per_question, created_at, updated_at.

3. questions

Stores multiple-choice questions associated with a course. Fields: id (PK), course_id (FK referencing courses.id), level, question, option1, option2, option3, option4, correct_answer, explanation, created_at, updated_at.

4. quiz_attempts

Records each student's quiz attempt including performance and anti-cheating metrics. Fields: id (PK), user_id (FK referencing users.id), course_id (FK referencing courses.id), score, total_questions, time_taken, seconds_taken, is_suspicious (boolean, default false), tab_switch_count (int, default 0), avg_time_per_question (float, nullable), started_at (timestamp, nullable), created_at, updated_at.

Relationships :

- users (teacher) to courses: One-to-Many relationship. A teacher can create multiple courses.
- users (student) to quiz_attempts: One-to-Many relationship. A student can have multiple quiz attempts.
- courses to questions: One-to-Many relationship. A course can contain multiple questions.
- courses to quiz_attempts: One-to-Many relationship. A course can have multiple quiz attempts from different students.

Implemented APIs

The backend exposes a comprehensive set of RESTful API routes to support all functionalities of the quiz system. These routes handle course management, question management, quiz operations, and result retrieval. The following table summarizes all implemented API endpoints, their HTTP methods, purposes, and current working status.

Method	Endpoint	Purpose	Status
GET	/teacher/courses	List all courses for logged-in teacher	Working
GET	/teacher/courses/create	Show course creation form	Working
POST	/teacher/courses	Create a new course	Working
GET	/teacher/courses/{id}/edit	Show edit form	Working
PUT	/teacher/courses/{id}	Update course	Working
DELETE	/teacher/courses/{id}	Delete course	Working
GET	/teacher/courses/{course_id}/questions	List questions for a course	Working
POST	/teacher/courses/{course_id}/questions	Add a new question	Working
GET	/teacher/questions/{id}/edit	Show edit question form	Working
PUT	/teacher/questions/{id}	Update question	Working
DELETE	/teacher/questions/{id}	Delete question	Working
GET	/teacher/suspicious-attempts	View suspicious attempts	Working
GET	/quiz/{course_id}	Show quiz page	Working
POST	/quiz/submit	Submit quiz (cheating detection logic implemented)	Working
GET	/result/{attempt_id}	Show result page	Working
GET	/leaderboard/{course_id}	Show leaderboard	Working
POST	/teacher/generate-ai-question	AI question generation (Gemini API integration - pending)	Pending

Note: The POST /teacher/generate-ai-question endpoint integrates with the Gemini API for AI-powered question generation. This feature is currently pending and is scheduled for completion in the next development sprint.

Authentication Workflow

The system implements a secure, role-based authentication workflow using Laravel Breeze with session-based authentication. The workflow is described step by step as follows.

1. **User Registration:** Users register through the /register route by providing their name, email address, and password. Upon successful registration, a new record is created in the users table.
2. **Role Assignment:** The is_teacher field in the users table determines the role of each user. A value of 1 designates the user as a Teacher, while a value of 0 designates the user as a Student. Role assignment is performed during or after registration.
3. **User Login:** Users authenticate through the /login route using their registered email and password. Laravel Breeze manages the session-based login mechanism, ensuring secure credential verification.
4. **Role-Based Middleware:** A custom TeacherMiddleware has been implemented to protect all routes under the /teacher/* prefix. This middleware checks whether the authenticated user has is_teacher = 1 before granting access to teacher-specific functionalities such as course and question management.
5. **Logout:** Users can terminate their session securely through the /logout route, which clears the session data and redirects users to the login page.

Current Development Progress

The development of the AI-Based Smart Quiz System is progressing according to the planned schedule. The majority of core backend and frontend features have been completed, with only the AI Question Generation feature pending implementation. The following table provides a detailed overview of the current development progress for each module.

Task	Progress (%)	Status
Project Setup (Laravel 12)	100%	Complete
Database Schema & Migrations	100%	Complete
Models & Relationships	100%	Complete
Authentication (Laravel Breeze)	100%	Complete
Teacher Middleware	100%	Complete

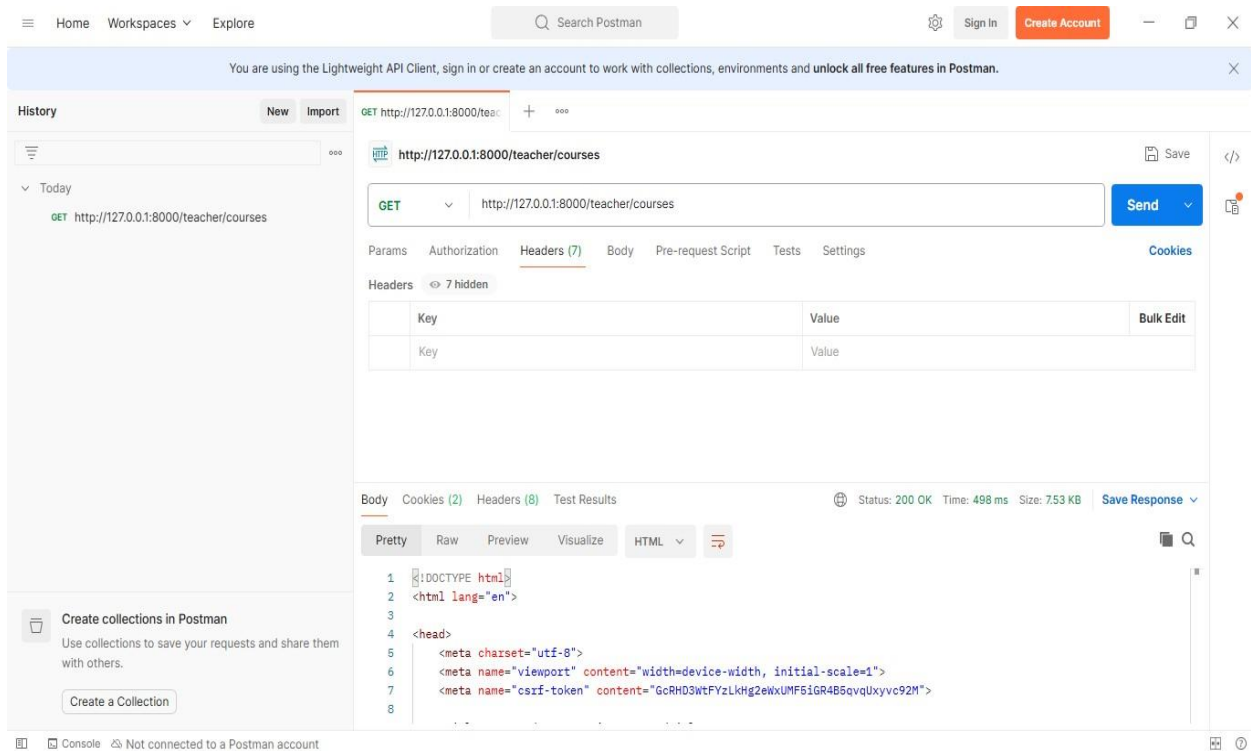
Course CRUD (Backend + Views)	100%	Complete
Question CRUD (Backend + Views)	100%	Complete
Quiz Page (UI)	100%	Complete
Quiz Submission (with Cheating Detection)	80%	In Progress
Result Page	100%	Complete
Leaderboard	100%	Complete
AI Question Generation	0%	Pending
Suspicious Attempts View	100%	Complete

The overall backend development is approximately 95% complete. The Quiz Submission module, which includes cheating detection logic, is currently at 80% and is being finalized. The AI Question Generation feature, which relies on integration with the Gemini API, is the only major feature pending and is scheduled for completion in the next development sprint. All other modules, including authentication, course management, question management, leaderboard, and suspicious attempts view, have been fully implemented.

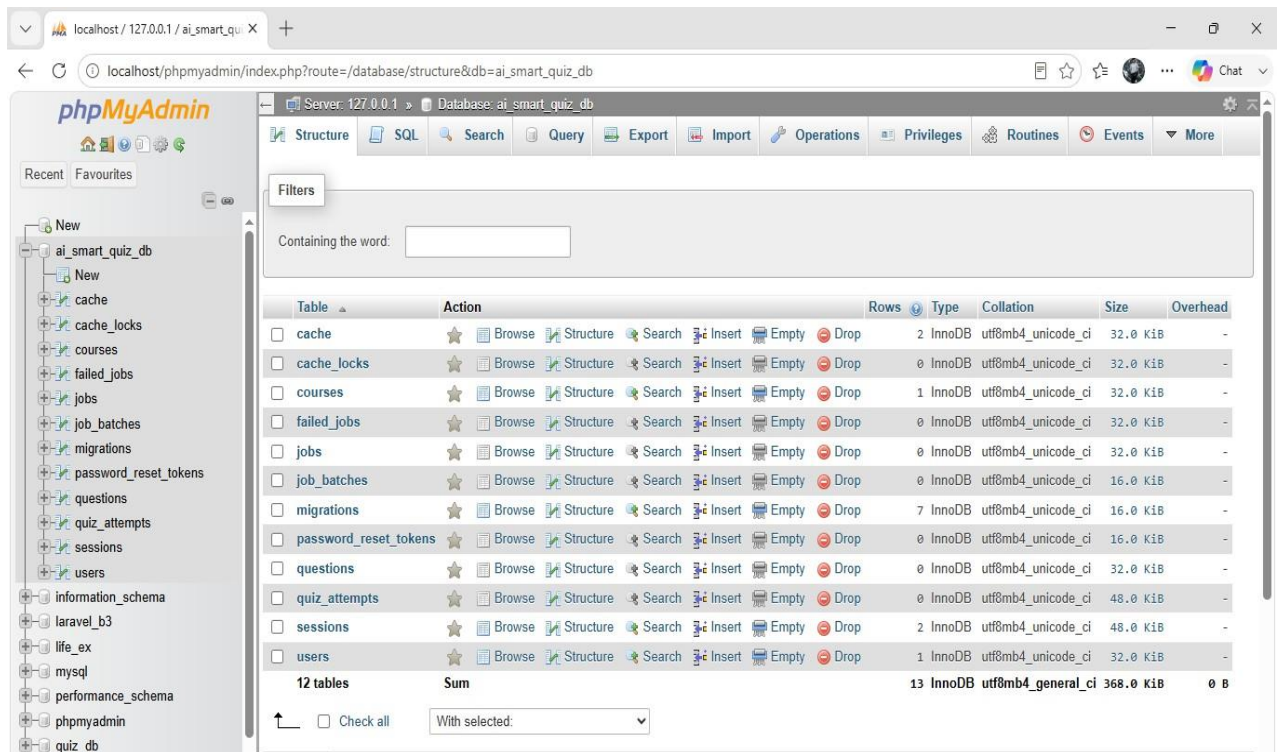
Screenshots of API Testing

The following screenshots provide visual evidence of the API testing conducted using Postman, the database structure as viewed in phpMyAdmin, the project folder structure, the teacher dashboard interface, and the GitHub repository for version control.

Screenshot 1: Postman API Test – GET /teacher/courses (200 OK Response) :



Screenshot 2: Database Tables (phpMyAdmin – ai_smart_quiz_db) :



localhost / 127.0.0.1 / ai_smart_quiz X

localhost/phpmyadmin/index.php?route=/table/structure&db=ai_smart_quiz_db&table=courses

Server: 127.0.0.1 > Database: ai_smart_quiz_db > Table: courses

Table structure Relation view

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 id	bigint(20)		UNSIGNED	No	None		AUTO_INCREMENT	Change Drop More
☐	2 teacher_id	bigint(20)		UNSIGNED	No	None			Change Drop More
☐	3 name	varchar(255)	utf8mb4_unicode_ci		No	None			Change Drop More
☐	4 description	text	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐	5 time_per_question	int(11)			No	30			Change Drop More
☐	6 created_at	timestamp			Yes	NULL			Change Drop More
☐	7 updated_at	timestamp			Yes	NULL			Change Drop More

☐ Check all With selected: [Browse](#) [Change](#) [Drop](#) [Primary](#) [Unique](#) [Index](#) [Spatial](#) [Fulltext](#)

[Add to central columns](#) [Remove from central columns](#)

[Print](#) [Propose table structure](#) [Track table](#) [Move columns](#) [Normalise](#)

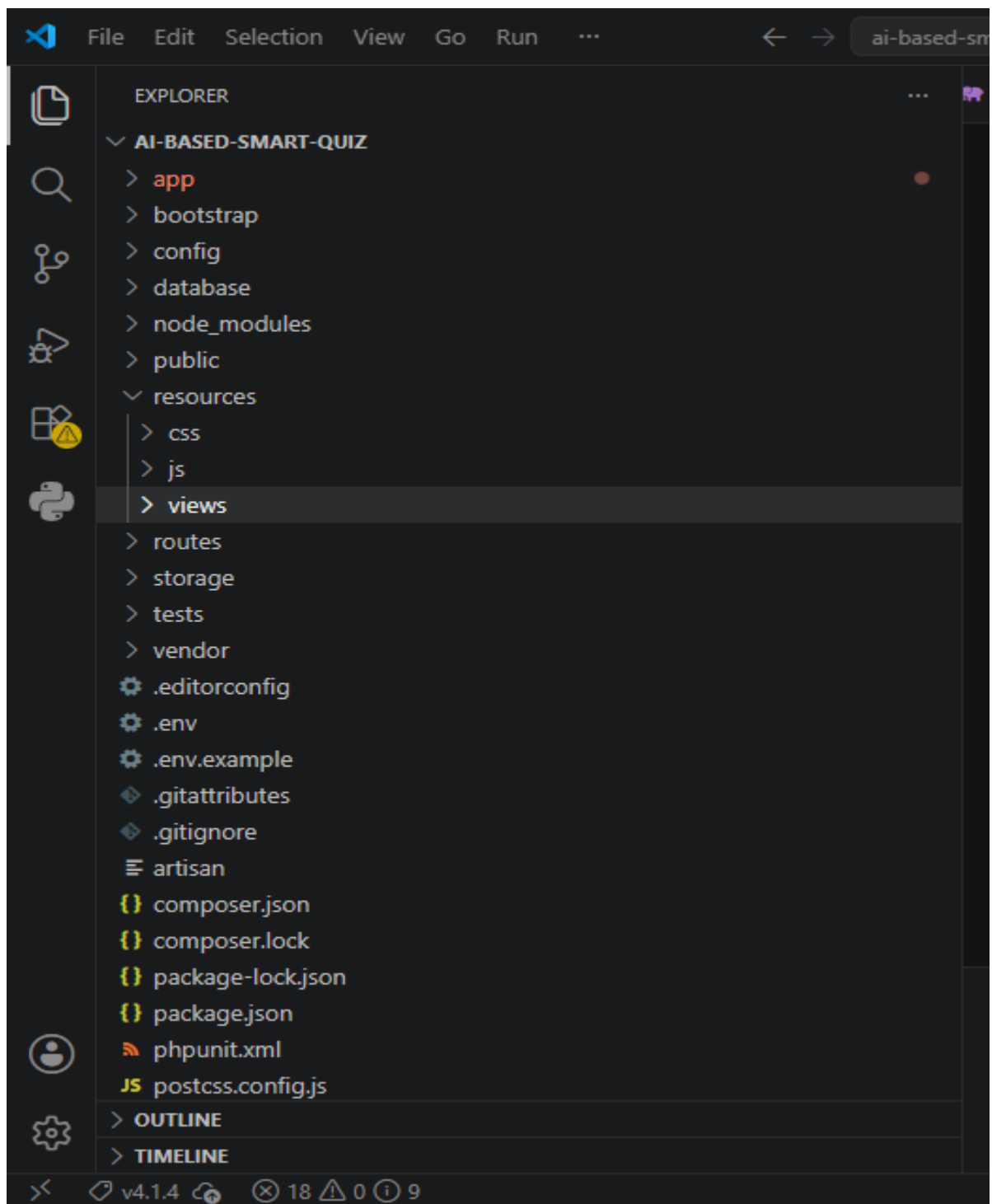
[Add](#) 1 column(s) after updated_at [Go](#)

Indexes

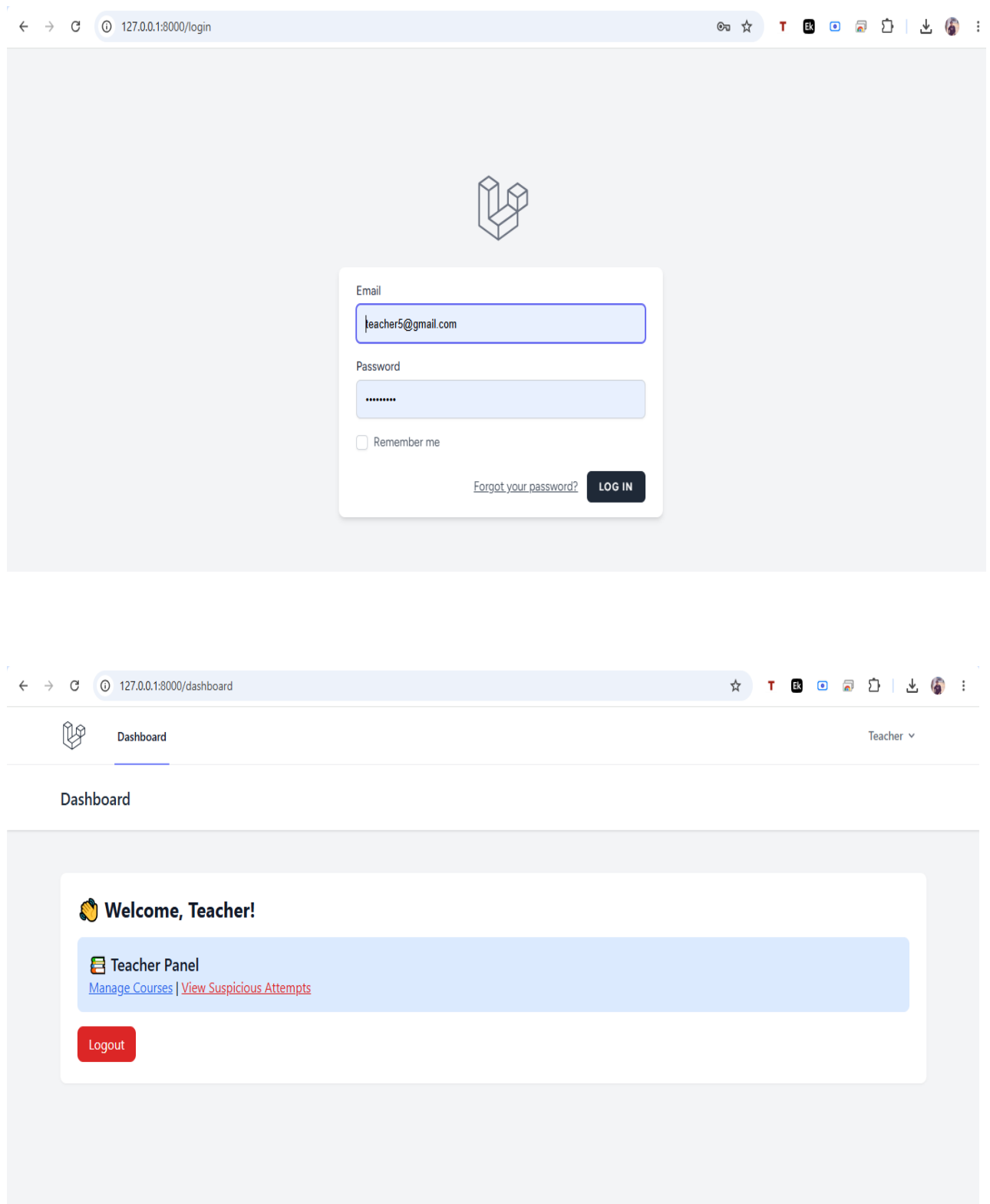
Action	Keyname	Type	Unique	Packed	Column	Cardinality	Collation	Null	Comment
Edit Rename Drop	PRIMARY	BTREE	Yes	No	id	1	A	No	
Edit Rename Drop	courses_teacher_id_foreign	BTREE	No	No	teacher_id	1	A	No	

Console

Screenshot 3: Project Folder Structure (VS Code) :



Screenshot 4: Authentication Process – User login successful, redirected to dashboard :



Screenshot 5: GitHub Repository :

GitHub Repository Link – <https://github.com/sudipto224/AI-based-Smart-Quiz-System>

github.com/sudipto224/AI-based-Smart-Quiz-System/blob/main/README.md

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- Week 03 Assignment
- backend
- database
- diagrams
- documentation
- postman
- README.md

AI-based-Smart-Quiz-System / README.md

sudipto224 Add repository structure section to README 970c8e8 · last week History

Preview Code Blame 146 lines (91 loc) · 4.83 KB

Raw Download Edit

AI-Based Smart Quiz System

Official Team Name: CSE4204-8A-T03

Team Information

Role	Full Name	Student ID
Team Leader + Backend	Sudipto Das	11220320830
Frontend	Chaon Das	11210320629
Backend Developer	Aditya Das	11220320830